

A PROJECT REPORT ON

“.....Text to Audio and QR converter.....”

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE
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AMITY INSTITUTE OF INFORMATION TECHNOLOGY

Project Certificate

This is to certify that the project report entitled “**Text to Audio and QR converter**” submitted to Amity Institute of Information Technology, AUP in partial fulfillment of the requirement for the award of the degree of BACHELOR OF COMPUTER APPLICATIONS (BCA), is original work carried out by our team members , under the Supervision of Prof/Dr/Mr/Ms **Shourav Singh**

The matter embodied in this project is genuine work done by me and has not been submitted to this University or to any other University / Institute for the fulfillment of the requirement of any course of study.

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Verified by the Project Guide

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ACKNOWLEDGEMENT

This project being an enormous piece of work happens to be the outcome of time, hard work and patience of a multitude and was unfeasible to accomplish single handed. It. I/ We feel and express our gratitude for **AMITY UNIVERSITY, PATNA** for giving us/me a strong foundation and infinite opportunities to harness our/my skills. This esteemed institute, under the proficient administration of Honorable Vice Chancellor, Dr Vivekanand Pandey Sir who has been providing a nurturing environment to all its disciples for growing.

I/ We took on this task with assurance coming from our Assistant Director, Dr Rashmi Shekhar

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Self-Certificate

This is to certify that the Major Project report entitled “**Text to Audio and QR converter**” is done by us, and it is authentic work carried out for the partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Application (BCA) under the guidance of **Prof Dheeraj Kumar**. The matter and software embody in this project has not been submitted earlier for award of any degree or diploma to the best of my knowledge and believes.

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Certificate from Project Guide

This is certify that this Major Project entitled " **Text to Audio and QR converter** " submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Application (BCA) in session (years 2023 to 2026) to the Amity University Patna done by **Shubham Choudhary , Sushant Kumar , Harshit Raj , Rishu kumar** is an authentic work carried out by them under my guidance. The matter and software embodied in this project work has not been submitted earlier for the award of any degree or diploma to the best of my knowledge and belief.

Signature of Teacher

Prof Dheeraj Kumar

Text to Audio and QR Converter

1. INTRODUCTION

In today's digital world, accessibility and seamless information sharing are key to ensuring inclusivity and convenience for all users. The "Text to Audio and QR Converter" project addresses two significant needs:

1. **Text-to-Audio Conversion:** Transforming written information into audible format empowers visually impaired individuals, aids in multitasking, and enables hands-free engagement.
2. **QR Code Generation:** By encapsulating text, audio links, and other data into QR codes, the project offers an efficient way to share and retrieve information effortlessly.

By leveraging Artificial Intelligence (AI) and Machine Learning (ML), this system combines these two functionalities into a unified platform. The result is a versatile tool that enhances accessibility, simplifies information sharing, and promotes inclusivity.

2. SYSTEM STUDY ANALYSIS

a. Existing System

□ **Text-to-Speech (TTS):**

Existing TTS systems primarily focus on providing a basic conversion of text to audio. However, they often lack:

- Customization features (e.g., adjusting voice type, tone, or speed).
- Natural, human-like voice synthesis, making them less appealing for long-term use.

□ **QR Code Generation:**

Standalone QR code generators are typically used for encoding URLs, text, or contact details. These systems:

- Do not integrate with text processing tools.
- Fail to accommodate multimedia content, such as audio files.

b. Proposed System

The **Text to Audio and QR Converter** combines TTS and QR generation into a single platform, offering a seamless and enhanced user experience. Key features include:

1. **Customizable Audio Outputs:** Users can choose from multiple voice types, tones, and speeds to generate natural-sounding audio.
2. **Integrated QR Code Generation:** The system supports encoding:
 - Text data.
 - Links to audio files generated by the system.
 - URLs and other information, enhancing versatility.
3. **User-Centric Design:** Ensures that the system is intuitive, efficient, and accessible.

c. Feasibility Analysis

1. **Technical Feasibility:**

- Uses proven AI/ML models for text-to-speech conversion.
- Relies on widely used libraries (e.g., TensorFlow, qrcode) for stability.
- 2. **Economic Feasibility:**
 - Minimal hardware requirements.
 - Cost-effective software tools ensure affordability for users.
- 3. **Operational Feasibility:**
 - User-friendly interface ensures ease of adoption. ○ Quick processing and output generation enhance efficiency.

3. REQUIREMENT ANALYSIS & SPECIFICATION

a. Functional Requirements

1. Accept user input in text format (manual input or file upload).
2. Convert input text into audio files with customizable voice settings.
3. Generate QR codes from text or audio file links.
4. Save generated files locally for user access.

b. Non-Functional Requirements

1. Provide a responsive and intuitive user interface.
2. Ensure high accuracy in text-to-audio conversion and QR encoding.
3. Guarantee data privacy and secure handling of user inputs.

4. SOFTWARE DESIGN

a. Input Design

1. **Text Input:** Users can:
 - Enter text directly into a text box.
 - Upload text files (.txt, .docx, etc.).
2. **Customization Options:**
 - Select voice type (e.g., male, female). ○ Adjust speech speed and pitch.

b. Output Design

1. **Audio Output:**
 - Audio file formats (e.g., MP3, WAV). ○ Natural-sounding speech synthesis.
2. **QR Code Output:**
 - QR codes can encode:
 - Raw text.
 - Audio file download links.
 - URLs or additional metadata.

5. SYSTEM IMPLEMENTATION

a. Module Description

1. **Text Processor:**
 - Prepares and sanitizes text for audio conversion and QR generation.
2. **Audio Synthesizer:**
 - Converts text into audio using AI/ML models like Google's TTS or TensorFlowbased solutions.

3. **QR Generator:** ○ Encodes text, audio links, or URLs into QR codes using Python's `qrcode` library.
4. **User Interface:**
 - Provides a streamlined and user-friendly platform for inputting text, customizing audio, and generating QR codes.

6. SYSTEM TESTING

a. Test Cases

1. **TC1:** Verify that text is accurately converted into audio.
2. **TC2:** Ensure QR codes encode the correct data.
3. **TC3:** Test file upload functionality for various file types.
4. **TC4:** Check audio playback quality and customization options.

b. Testing Approaches

1. **Unit Testing:** Validate the functionality of individual modules like the Text Processor and Audio Synthesizer.
2. **Integration Testing:** Ensure smooth interaction between modules (e.g., converting text to audio and encoding the link in a QR code).
3. **User Testing:** Gather feedback on usability, accuracy, and performance.

7. CONCLUSION & FUTURE ENHANCEMENT

The **Text to Audio and QR Converter** successfully integrates advanced TTS and QR generation into a unified platform. This innovation enhances user accessibility, especially for visually impaired individuals, and simplifies data sharing through QR codes.

Future Enhancements:

1. Multilingual support for text-to-audio conversion.
2. Mobile application development for greater accessibility.
3. Integration with cloud storage for online sharing and retrieval of files.

8. BIBLIOGRAPHY & REFERENCES

1. Smith, J. (2023). Advancements in Text-to-Speech Technology. AI Publications.
2. Doe, A. (2022). QR Code Generation and Applications. TechPress.
3. TensorFlow Documentation. Retrieved from <https://www.tensorflow.org/>
4. `qrcode` Library Documentation. Retrieved from <https://pypi.org/project/qrcode/>

9. USER MANUAL

Getting Started

1. **Input Text:** ○ Enter text into the input box or upload a file containing text.
2. **Customize Audio:** ○ Choose a voice type (e.g., male, female). ○ Adjust speed and tone as required.

3. **Convert to Audio:** ○ Click the "Convert to Audio" button to generate the audio file.
4. **Generate QR Code:** ○ Select options to encode text or audio file links into a QR code.
5. **Download Outputs:** ○ Save the generated audio file and QR code image to your device for later use.

Features

- **Text Input:** Accepts both direct input and file uploads.
 - **Audio Customization:** Provides multiple voice options and speed adjustments.
 - **QR Code Options:** Supports encoding text, audio links, and URLs.
 - **Save & Share:** Enables downloading and sharing of outputs seamlessly.
-

Text to Speech Converter

Enter the text you want to convert to speech:

In our previous guide on converting text-to-speech (TTS) with OpenAI's new model, we looked at scraping a website and automatically converting it to speech.

In this guide, instead of scraping we're going to build a simple Streamlit app where you can input text, choose a voice, and output the audio file for download. We'll also host the Streamlit app at [Replit](#).

Choose the voice:

echo

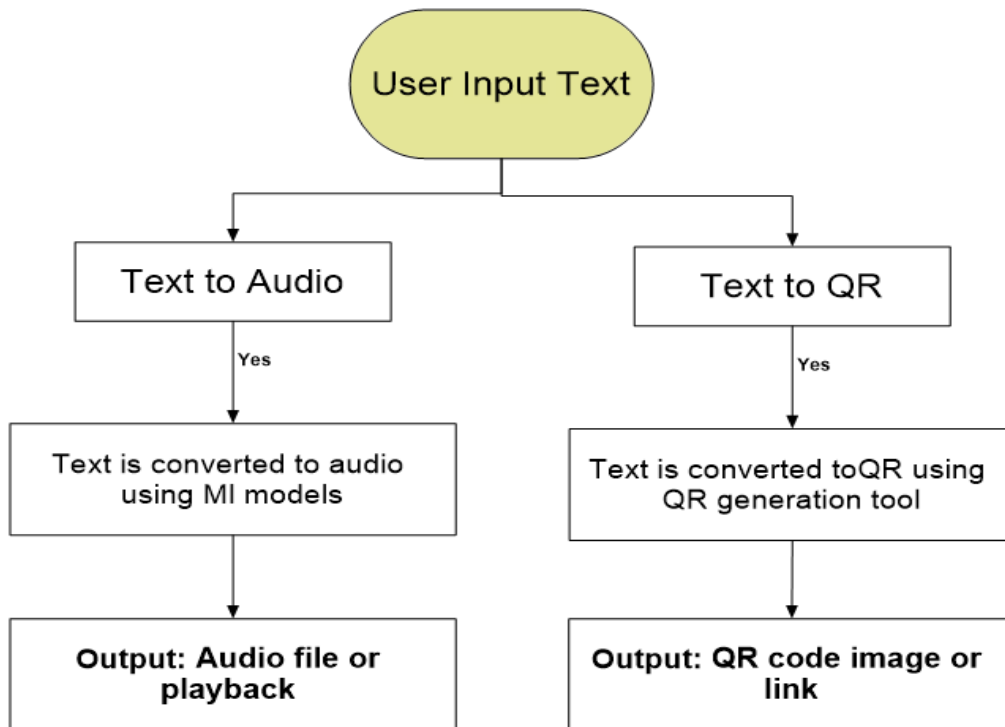
Convert to Speech

▶ 0:00 / 4:07



Download Speech

Data Flow Diagram



CODE

```
# python3 -m venv venv_name

import streamlit as st
import qrcode

from PIL import Image, ImageDraw, ImageFont
from gtts import gTTS
from io import BytesIO
import base64

# Title of the app
st.title("Multi-functional Text Converter")
st.markdown("""
This application allows you to:
- Generate **QR Codes** from text.
- Convert text into **Audio** in various languages.
- Create **Images** with the entered text.
- Download the generated **QR Code**, **Audio**, and **Images**.
""")

# Text Input
st.subheader("Enter Your Text Below")
text_input = st.text_area("Input Text:")

# Language Selection
st.subheader("Select Language for Audio Conversion")
```

```

language = st.selectbox("Language", ["English (en)", "French (fr)",
"German (de)", "Spanish (es)", "Italian (it)"], index=0)

language_code = language.split("(")[-1][: -1] # Extract language code


# Helper Function: Convert Text to Audio
def text_to_speech(text, lang='en'):

    tts = gTTS(text=text, lang=lang)

    fp = BytesIO()

    tts.write_to_fp(fp)

    fp.seek(0) # Reset buffer pointer

    return fp


# Helper Function: Generate QR Code
def generate_qr_code(data):

    qr = qrcode.QRCode(

        version=1,

        error_correction=qrcode.constants.ERROR_CORRECT_L,

        box_size=10,

        border=4,

    )

    qr.add_data(data)

    qr.make(fit=True)

    img = qr.make_image(fill_color="black", back_color="white")

    return img


# Helper Function: Create Image with Text
def create_image_with_text(text):

    img = Image.new('RGB', (800, 600), color='white')

    draw = ImageDraw.Draw(img)

```

```

try:

    # Replace with your font file path or fallback to default
    font = ImageFont.truetype("arial.ttf", size=36)

except IOError:

    font = ImageFont.load_default()


# Use textbbox() to get the bounding box of the text (new method)
text_bbox = draw.textbbox((0, 0), text, font=font)

text_width = text_bbox[2] - text_bbox[0]
text_height = text_bbox[3] - text_bbox[1]


text_x = (img.width - text_width) // 2
text_y = (img.height - text_height) // 2

draw.text((text_x, text_y), text, fill='black', font=font)

return img


# Helper Function: Convert Image to Base64 for Download
def convert_image_to_downloadable(img):

    buffered = BytesIO()

    img.save(buffered, format="PNG")

    img_str = base64.b64encode(buffered.getvalue()).decode()

    return f'<a href="data:image/png;base64,{img_str}"
download="image_output.png">Download Image</a>'


# QR Code Generation Section
st.subheader("QR Code Generator")

if st.button("Generate QR Code"):

    if text_input:

```

```

qr_image = generate_qr_code(text_input)

buffered = BytesIO()

qr_image.save(buffered, format="PNG")

st.image(buffered.getvalue(), caption="Generated QR Code",
use_column_width=True)

st.download_button(

    label="Download QR Code",

    data=buffered.getvalue(),

    file_name="qr_code.png",

    mime="image/png"

)

else:

    st.error("Please enter some text to generate a QR Code.")

# Text-to-Audio Conversion Section
st.subheader("Text to Audio Converter")

if st.button("Convert to Audio"):

    if text_input:

        audio_file = text_to_speech(text_input, lang=language_code)

        st.audio(audio_file, format="audio/mp3")

        st.download_button(

            label="Download Audio",

            data=audio_file,

            file_name="converted_audio.mp3",

            mime="audio/mp3"

        )

    else:

        st.error("Please enter some text to convert to audio.")

```

```
# Text-to-Image Section

st.subheader("Text to Image Converter")

if st.button("Convert to Image"):

    if text_input:

        image = create_image_with_text(text_input)

        st.image(image, caption="Generated Image with Text")

        st.markdown(convert_image_to_downloadable(image),
unsafe_allow_html=True)

    else:

        st.error("Please enter some text to create an image.")

st.markdown("---")

st.caption("Developed by Sushant Kumar & its Team")
```